Covariate-adjusted statistical dependence representation through partial copulas: bounds and new insights

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Abstract

In this paper, we revisit the notion of partial copula, originally introduced to test conditional independence, highlighting its capability to represent the dependence between two random variables after removing their dependence with a covariate. Building upon results previously presented in the literature, we show that partial copulas can be seen as a nonlinear analogue of partial correlation. Then, we prove several results showing how dependence properties of the conditional copulas constrain the form of the partial copula. Finally, a simulation study is conducted to illustrate the results and to show the potential of the partial copula as a way to describe covariate-adjusted statistical dependence. This highlights the potential of the method to be used in causal inference problems and to recover the true sign of a causal effect.

1 Introduction

Quantifying the strength of the statistical association between two random variables, controlled by one or more variables (usually called covariates or control variables), is an old problem in statistics and widely discussed in the context of linear models and regression coefficients [Kutner et al., 2005, Baba et al., 2004, Kim, 2015]. It is well established in the statistical literature that, under wide conditions, conditionally independent random variables are unconditionally correlated [see theorem 2 in O'Neill, 2009]. In fact, latent variables ensuring conditional independence are a common tool in statistics to "create" or "explain" dependence between variables - see, for instance, factor models [Joe, 2014, Van Bork et al., 2017, Nguyen et al., 2020] and de Finetti's representation theorem [O'Neill, 2009]. Therefore, in the context of causal models, two variables causally unrelated may be correlated if they are both affected by a third variable, called confounder [Pearl, 2009, Pearl et al., 2016]. For this reason, the analysis of conditional and partial associations is of great importance in statistical analysis of observational data; in the context of regression analysis, this is done by adding control variables [Kutner et al., 2005], and, in the context of correlation analysis, this is done through partial correlation [Baba et al., 2004, Kim, 2015]. A brief introduction on how to choose control variables properly (from a causal viewpoint) is presented in Cinelli et al. [2022].

In the same way that using a single numerical index (such as the expectation) is not as informative as knowing the distribution of a variable, it is also true that correlation coefficients fail to capture several aspects of the dependence structure of a random vector - for this reason, copula theory offers a much richer way to study statistical dependence. Copulas are functions such that, given the copula and the univariate (marginal) distributions of a random vector, the joint distribution of the variables is fully characterized - therefore, they fully characterize the dependence structure behind a random vector [Nelsen, 2006, Joe, 2014]. In fact, it has been proved that copulas offer sufficient and necessary conditions for independence [Theorem 2.4.2 in Nelsen, 2006] and conditional independence [Theorem 1 in González-López and Justus, 2024]. Embrechts et al. [2002] presents an example of two bivariate distributions that share the same marginal distributions and same Pearson's correlation value, but whose tail dependence is not the same because the copulas are not the

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same. This exemplifies the limitation of using Pearson's correlation coefficient to summarize the dependence structure behind a random vector. Therefore, having the notion of partial copula is of great relevance to fully study the stochastic dependence between two variables after controlling for some confounder, because they are a natural extension of the notion of partial correlation.

Partial copulas were introduced in Bergsma [2004] as a tool to construct a conditional independence test [which has also been explored in Bergsma, 2011; Patra et al., 2016; and Petersen and Hansen, 2021], and their use in vine copula and stochastic process modeling has been discussed in the literature [Bladt and McNeil, 2022, Bauer and Czado, 2016, Spanhel and Kurz, 2019]. However, to the best of our knowledge, the use of this notion as a way to formally study covariate-adjusted dependence has not been explored enough in the literature [despite the existence of some references exploring related matters, see Gijbels et al., 2015, and Spanhel and Kurz [2016]]. In the context of linear models, the use of copulas to enable causal inference was proposed in Park and Gupta [2012] and further studied in Falkenström et al. [2023]; the method proposed by the authors is a data-driven method that attempts to remove the confounder effect without measuring the confounding variable directly (which may require strong untestable assumptions). More recently, the use of vine copulas in causal inference was considered by Justus [2024] and Czado [2025] - however, to the best of our knowledge, the use of partial copulas to investigate causal effects is not yet properly explored in the literature.

Our main contributions are twofold. First, we give a practical interpretation of the notion of partial copula as a nonlinear analogue of partial correlation, suggesting that its use should not be limited to conditional independence testing. Second, we prove several results showing how properties of the conditional copulas - such as Kolmogorov distance dependence, quadrant positive dependence and Spearman's correlation - constrain the form of the partial copula. Additionally, the results are illustrated through simulations.

The paper is organized as follows. Next section provides all the necessary background to the study of partial copulas: Rosenblatt transformations, probability integral transform and copula functions are introduced. Section 3 is divided in three subsections, the first two discussing the interpretation of partial copula and the last one introducing new results connecting conditional and partial copulas. A simulation study is conducted in Section 4, and final considerations are given in Section 5.

2 Background

2.1 Copulas

Copulas are multivariate functions $[0,1]^d \to [0,1]$ used to describe the dependence structure of random vectors of dimension $d \in \mathbb{N}$. The definition in the bivariate case is presented below.

Definition 1. *A 2-copula is a function $C : [0,1]^2 \to [0,1]$ with the following properties:*

(i) $C(u,0) = C(0,v) = 0$, and $C(u,1) = u$, $C(1,v) = v$, $\forall u, v : u, v \in [0,1]$;

(ii) for every $u_i, v_i \in [0,1]$, $i = 1,2$, such that $u_1 \leq u_2$ and $v_1 \leq v_2$,

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) \geq 0.$$

Note that Definition 1 is equivalent to the definition of 2-variate cumulative distribution function (CDF) whose marginal distributions are uniform in $[0,1]$ range. Therefore, given any two random variables U, V such that $U \sim U(0,1)$ and $V \sim U(0,1)$, the cumulative distribution function $P(U \leq u, V \leq v)$ is necessarily a copula (because it satisfies Definition 1).

Sklar's theorem [Sklar, 1959, Nelsen, 2006] guarantees that, given any two random variables X and Y, it holds that

$$P(X \leq x, Y \leq y) = C(F_X(x), F_Y(y)),$$

where F_X and F_Y are, respectively, the cumulative distribution functions of X and Y, and C is a copula - if the variables are continuous, then C is unique and it may be called "the copula of (X,Y)". It's possible to note that the copula of (X,Y) is the CDF of $(F_X(X), F_Y(Y))$.

The representation provided by Sklar's theorem shows that copulas are a "marginal-free" way to study the dependence between two (or more) random variables, which makes clear that this is a central notion in the study of stochastic dependence. For further arguments regarding the importance of copula theory in the study of stochastic dependence, see Embrechts et al. [2002] and see also the monographs from Nelsen [2006] and Joe [2014].

2.2 Probability integral transform and Rosenblatt transforms

An important result in both copula theory and computational statistics is the so-called "probability integral transform" [Angus, 1994], presented below. This result ensures that a cumulative distribution function makes a variable uniformly distributed when it is used as a transformation applied in the random variable. Given that copulas are cumulative distribution functions of variables uniformly distributed in $[0, 1]$, the importance of this result for copula theory seems evident.

Proposition 1 (Probability Integral Transform)**.** *Let F be an absolutely continuous cumulative distribution function, and let F^{-1} be the inverse function of F. Then,*

- *If $U \sim U(0, 1)$, then $F^{-1}(U)$ is a random variable with cumulative distribution function F (which will be denoted as $F^{-1}(U) \sim F$).*
- *Conversely, if $X \sim F$, then $F(X)$ is a random variable with uniform distribution on $(0, 1)$, that is, $F(X) \sim U(0, 1)$.*

An even more general result is available in Rosenblatt [1952]: given a sequence of random variables $X_1, \dots, X_n$, the random variables

$$F_1(X_1),\ F_{X_2|X_1}(X_2|X_1),\ \dots,\ F_{X_n|X_1,\dots,X_{n-1}}(X_n|X_1, \dots, X_{n-1})$$

are independent and uniformly distributed in the $[0, 1]$ range. These functions are usually called "Rosenblatt transforms" and they can be seen as a kind of generalization of the probability integral transform to the multivariate case. In statistical literature, those transformations have already been employed to test conditional independence [Bergsma, 2004, Song, 2009, Patra et al., 2016] and, more recently, to construct a nonparametric notion of error term in regression models [Patra et al., 2016]. A central result connecting Rosenblatt transforms and conditional independence is given by next proposition.

Proposition 2. *[Patra et al., 2016] Let (X, Y, Z) be an absolutely continuous random vector. Therefore, $X \perp\!\!\!\perp Y|Z$ if and only if $F_{X|Z}(X|Z)$, $F_{Y|Z}(Y|Z)$ and Z are mutually independent.*

This result suggests that studying the joint distribution of $F_{X|Z}(X|Z)$ and $F_{Y|Z}(Y|Z)$ is a natural way to study the dependence between X and Y adjusted by Z, because those two quantities would be independent in the particular case that they are conditionally independent given Z (therefore, this joint distribution would remove the "spurious association" induced by Z). In the context of causal inference, if Z is the confounder of (X, Y), the two variables would be conditionally independent given Z, provided they are causally unrelated (see considerations about fork models in [Pearl, 2009, Pearl et al., 2016].

3 The partial copula

3.1 Definition and interpretation

This section presents the notion of partial copula, originally introduced in Bergsma [2004].

Definition 2. *Let (X, Y, Z) be an absolutely continuous random vector. The partial copula of (X, Y) controlled by Z, which we denote by $C_{X,Y;Z}$, is defined as*

$$C_{X,Y;Z}(x, y) = P(U_X \leq x, U_Y \leq y),$$

where $U_X = F_{X|Z}(X|Z)$ and $U_Y = F_{Y|Z}(Y|Z)$.

Remark 1. *Note that $C_{X,Y;Z}$ fully satisfies the definition of copula (Definition 1), because $C_{X,Y;Z}$ is the cumulative distribution function of two random variables uniformly distributed in* $[0, 1]$.

The partial copula $C_{X,Y;Z}$ may be seen - in a similar way to partial correlation - as a notion that represents the stochastic dependence between (X, Y) after removing the effect of a covariate Z. Further, differently from the notion of conditional copula, $C_{X,Y;Z}$ does not depend on the specification of a specific value z of Z.

Next result shows that the partial copula $C_{X,Y;Z}$ can be written as a mixture of all conditional copulas $C_{X,Y|Z=z}$; this result, in addition to providing an analytic representation for the partial copula, highlights the capability of the concept in capturing conditional dependence.

Proposition 3. *Let (X, Y, Z) be a continuous random vector. The partial copula $C_{X,Y;Z}$ can be written as*

$$C_{X,Y;Z}(x, y) = \int_{-\infty}^{\infty} C_{X,Y|Z=z}(x, y) f_Z(z) dz,$$

where $C_{X,Y|Z=z}$ is the conditional copula of (X, Y) given $Z = z$.

Proof. Note that

$$\begin{aligned}
C_{X,Y;Z}(x, y) &= P(U_X \leq x, U_Y \leq y) \\
&= \int_{-\infty}^{\infty} P(U_X \leq x, U_Y \leq y | Z = z) f_Z(z) dz \\
&= \int_{-\infty}^{\infty} P(F_{X|Z}(X|Z) \leq x, F_{Y|Z}(Y|Z) \leq y | Z = z) f_Z(z) dz \\
&= \int_{-\infty}^{\infty} P(F_{X|Z}(X|z) \leq x, F_{Y|Z}(Y|z) \leq y | Z = z) f_Z(z) dz \\
&= \int_{-\infty}^{\infty} C_{X,Y|Z=z}(x, y) f_Z(z) dz,
\end{aligned}$$

where the fourth equality comes from applying the substitution principle for conditional distributions [Proposition 4.1 of James, 1981]; and the last equality comes from applying Sklar's theorem [Sklar, 1959] in $P(F_{X|Z}(X|z) \leq x, F_{Y|Z}(Y|z) \leq y|Z = z)$.

This application of Sklar's theorem is justified because $P(F_{X|Z}(X|z) \leq x, F_{Y|Z}(Y|z) \leq y|Z = z)$ is a cumulative distribution function for any $z \in \mathbb{R}$; Furthermore, $F_{X|Z}(X|z)$ and $F_{Y|Z}(Y|z)$ are, respectively, strictly increasing transformations of x and y, because they are cumulative distribution functions for any $z \in \mathbb{R}$; then, from Theorem 2.4.3 in [Nelsen, 2006], the conditional copula identified by Sklar's theorem in $P(F_{X|Z}(X|z) \leq x, F_{Y|Z}(Y|z) \leq y|Z = z)$ is the same as the conditional copula of $P(X \leq x, Y \leq y|Z = z)$. Therefore, the last equality is proved taking into account that the univariate conditional distributions are uniform in $[0, 1]$. □

Proof of Proposition 3 provides a formal justification for a similar expression that appears in proof of Proposition 1 in Gijbels et al. [2015].

Remark 2. *A straightforward adaptation of the proof of Proposition 3 [see also Equation 2.1 in Joe, 1996] shows that, given a continuous random vector (X, Y, Z),*

$$\begin{aligned}
P(X \leq x, Y \leq y) &= \int_{-\infty}^{\infty} P(X \leq x, Y \leq y | Z = z) f_Z(z) dz \\
&= \int_{-\infty}^{\infty} C_{X,Y|Z=z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)) f_Z(z) dz.
\end{aligned}$$

Therefore, the partial copula can also be seen as the distribution of (X, Y) under the assumption that each variable is independent of Z but keeping the same conditional copulas as in the true distribution of (X, Y, Z).

An interesting consequence of the last proposition - summarized in the following corollary - is that, when the "simplifying assumption" [commonly used in Vine copula modeling, see Czado and Nagler, 2022] is adopted, then the partial copula coincides with the conditional copula.

Corollary 1. *Let (X, Y, Z) be a continuous random vector. If the conditional copula of (X, Y) given $Z = z$ is constant as a function of z ("simplifying assumption"), then*

$$C_{X,Y;Z}(x, y) = C_{X,Y|Z}(x, y).$$

Proof. This result is a direct consequence of integrating $C_{X,Y|Z=z}(x, y)f_Z(z)$ with respect to z and taking into account that, from the theorem assumption, $C_{X,Y|Z=z}(x, y)$ is constant as a function of z.

$$\begin{aligned}
C_{X,Y;Z}(x, y) &= \int_{-\infty}^{\infty} C_{X,Y|Z=z}(x, y)f_Z(z)dz \\
&= \int_{-\infty}^{\infty} C_{X,Y|Z}(x, y)f_Z(z)dz \\
&= C_{X,Y|Z}(x, y)\int_{-\infty}^{\infty} f_Z(z)dz \\
&= C_{X,Y|Z}(x, y).
\end{aligned}$$

□

From this result, it also becomes clear that when (X, Y) are conditionally independent given Z (which produces the conditional copula $C_{X,Y|Z=z}(x, y) = xy$), then the partial copula is the independence copula, which is coherent with lemma 3.1 of Patra et al. [2016]. However, the reciprocal statement is not true - in the next remark, we construct a counterexample based on a model that commonly appears in the literature as a counterexample for certain statistical claims concerning pairwise independence or conditional independence [e.g., Nelsen and Úbeda-Flores, 2012, González-López and Justus, 2024].

Remark 3. *Corollary 1 entails that, if $X \perp\!\!\!\perp Y|Z$, then $C_{X,Y;Z}(x, y) = xy$ (or, equivalently, U_X and U_Y are independent).*

On the other hand, if U_X and U_Y are independent, it is not necessarily true that $X \perp\!\!\!\perp Y|Z$. Consider the following example: if $Z \sim U(0, 1)$ and $C_{X,Y|Z=z}(x, y) = xy + (2z - 1)xy(1 - x)(1 - y)$ (that is, a Farlie-Gumbel-Morgenstern copula with parameter $2z - 1$), then from Proposition 3,

$$C_{X,Y;Z}(x, y) = \int_0^1 [xy + (1 - 2z)xy(1 - x)(1 - y)]dz = xy,$$

which is the independence copula, which entails, from the definition of the partial copula, that U_X and U_Y are independent. However, except for $Z = 0.5$, X and Y are not conditionally independent given $Z = z$. Note that if $X \perp\!\!\!\perp Z$, $Y \perp\!\!\!\perp Z$ and $X, Y \sim U(0, 1)$, then this example produces a trivariate Farlie-Gumbel-Morgenstern copula .

Therefore, $U_X \perp\!\!\!\perp U_Y$ is a necessary but not sufficient condition for $X \perp\!\!\!\perp Y|Z$. A possible explanation for this fact is that, in the example presented in the remark, the conditional association of (X, Y) given $Z = z$ is positive for some values of z and negative for other values of z. In this sense, the partial copula represents a kind of "average conditional association"; this fact is shown in next remark.

Remark 4. *Let $g(x, y, z) = C_{X,Y|Z=z}(x, y)$. $g(x, y, Z)$ is a random variable, and Proposition 3 entails that*

$$C_{X,Y;Z}(x, y) = E[g(x, y, Z)],$$

which means that the partial copula is the expected conditional copula.

3.2 Partial copula as an extension of partial correlation

Next remark underscores the connection between the partial copula and the notion of partial correlation.

Remark 5. *A possible way to define the partial correlation between two random variables X and Y, given Z, is to construct two regression models $X = \alpha + \beta Z + \epsilon_{X;Z}$ and $Y = \gamma + \theta Z + \epsilon_{Y;Z}$ and then compute the Pearson's correlation between the error terms $\epsilon_{X;Z}$ and $\epsilon_{Y;Z}$ [Kim, 2015].*

More recently, Patra et al. [2016] studied the problem of how to construct a nonparametric notion of error term, that is, a notion of error that does not assume a specific functional form for the relationship between the regressor and the response variable. Imposing some conditions that are reasonable to expect from a notion of error term (such as being independent from the regressor), Patra et al. [2016] showed that the Rosenblatt transforms $F_{X|Z}(X|Z)$ and $F_{Y|Z}(Y|Z)$ satisfy those conditions and, therefore, may be seen as a general notion of error term. [for further considerations, see also Section 4 of Justus, 2024]

Partial copulas are the joint CDF of the nonparametric error terms $F_{X|Z}(X|Z)$ and $F_{Y|Z}(Y|Z)$, and, therefore, can be seen as a nonparametric extension of the notion of partial correlation.

Previous results in the literature have established that zero partial correlation is not equivalent to conditional independence, except in some special cases such as the multivariate normal distribution [Baba et al., 2004]. Next example shows that, even under conditional independence, the partial correlation may be arbitrarily close to 1, which suggests that the partial correlation may be a very poor measure of covariate-adjusted association.

Example 1. *Let $Z \sim N(0,1)$. Let $X|Z = z \sim N(z^2, \sigma^2)$ and $Y|Z = z \sim N(z^2, \sigma^2)$ be conditionally independent given $Z = z$. It is straightforward to see that, in this case,*

$$Corr(X,Z) = Corr(Y,Z) = 0,$$

which entails, from the definition of partial correlation, that $Corr(X,Y;Z) = Corr(X,Y)$. On the other hand, X and Y are exchangeable and, from Theorem 2 in O'Neill [2009], it holds that

$$Corr(X,Y) = \frac{Var(E(X|Z))}{Var(E(X|Z)) + E(Var(X|Z))} = \frac{Var(Z^2)}{Var(Z^2) + \sigma^2}.$$

Therefore,

$$Corr(X,Y;Z) = Corr(X,Y) = \frac{2}{2+\sigma^2} > 0,$$

which becomes arbitrarily close to 1 as σ^2 approaches 0 - that is, $\lim_{\sigma^2 \to 0} Corr(X,Y;Z) = 1$ -, even though X and Y are conditionally independent given Z.

The result seen in Example 1 may look worrying: even when two variables are conditionally independent given a covariate, their partial correlation may be close to 1. An intuitive explanation for this fact may come from Remark 5: given that the partial correlation is the Pearson's correlation between the error terms of two linear regression models, only the "linear effect" of Z on the variables is being "removed". In Example 1, the expectation of each variable given Z is not linear (not even monotonic), and, therefore, the partial correlation fails in removing the effect of Z on the variables. On the other hand, if X and Y are conditionally independent given Z, then the partial copula is always the independence copula. In this sense, the partial copula fully removes the "spurious association" induced by Z. Therefore, except when a researcher has strong reasons to make some parametric assumptions (such as linearity), the partial copula looks to be a much more appropriate notion for representing covariate-adjusted statistical association.

3.3 Effect of properties of the conditional copulas on the partial copula

Given the functional relationship between the partial and the conditional copulas, an interesting research line that emerges is studying to what extent properties of the conditional copulas constrain the properties of the partial copula. A first result on this matter is given by next proposition. The result is based on the notions of quadrant positive and quadrant negative dependence: a copula C is said to have quadrant positive dependence (QPD) if $C(u,v) \geq uv$ for all $(u,v) \in [0,1]^2$. In an analogous way, the copula has quadrant negative dependence (QND) if $C(u,v) \leq uv$ for all $(u,v) \in [0,1]^2$ [Joe, 2014]. A direct consequence of the fact that the partial copula is an expected conditional copula is given by the following proposition.

Proposition 4. *Let (X,Y,Z) be a continuous random vector. If the conditional copula of (X,Y) given $Z = z$ has positive quadrant dependence for all $z \in \mathbb{R}$, then the partial copula $C_{X,Y;Z}(x,y)$ has positive quadrant dependence.*

In the same way, if the conditional copula of (X,Y) given $Z = z$ has negative quadrant dependence for all $z \in \mathbb{R}$, then the partial copula $C_{X,Y;Z}(x,y)$ has negative quadrant dependence.

Proof. If $g(x,y,z) = C_{X,Y|Z=z}(x,y) \geq xy$ for all $z \in \mathbb{R}$, then, by interpreting the constant xy as a degenerate random variable and applying the monotonicity of expectation [property E2 of section 3.3 of James, 1981], we have the inequality

$$xy \leq E[g(x,y,Z)] = C_{X,Y;Z}(x,y).$$

Therefore, $C_{X,Y;Z}(x,y) \geq xy$, as we wished to prove. The same argument holds for negative quadrant dependence. $\square$

An intuitive interpretation for Proposition 4 is that, whenever the conditional association between X and Y is positive [negative] for all values of Z, the partial copula also exhibits positive [negative] dependence.

Next proposition uses a dependence measure based on Kolmogorov distance, which, for simplicity, we will refer to it as Kolmogorov distance dependence (KDD). This coefficient was introduced in Wolff [1977] and Schweizer and Wolff [1981], and is defined as

$$D(X,Y) = 4 \sup_{x,y\in[0,1]} (|C_{X,Y}(x,y) - xy|).$$

Note that the behavior of KDD differs from the behavior of the so-called "concordance measures", such as Spearman's correlation coefficient, which may be equal to zero even if the two variables are not independent [García-Portugués, 2025]. This happens intuitively because dependence and concordance measures have two different goals: the former ones measure deviation from independence, and the last ones measure the degree of monotonic association [see Nelsen, 2006, Joe, 2014].

It is immediate to see that KDD is proportional to the Kolmogorov distance between a given copula and the independence copula; the constant 4 ensures that $D(X,Y) \in [0,1]$. This coefficient is constructed intending to measure in what extent a copula deviates from independence, without distinguishing between "positive" and "negative" dependence; in this sense, values closer to 0 represent independence and cases closer to 1 represent strong dependence - for further details, see Wolff [1977] and Schweizer and Wolff [1981]. The following proposition shows how the value of KDD in the conditional copulas constrains the value of KDD in the partial copula.

Proposition 5. *Let X, Y, Z be jointly continuous random variables. Then,*

$$D(U_X, U_Y) \leq \sup_{z\in\mathbb{R}} (D(X,Y|Z=z)).$$

Proof. Suppose $\sup_{z\in\mathbb{R}}(D(X,Y|Z=z)) = k$, for some $k \in [0,1]$. This means that, for every $z \in \mathbb{R}$, it holds that

$$k \geq D(X,Y|Z=z) = 4 \sup_{x,y\in[0,1]} (|C_{X,Y|Z=z}(x,y) - xy|),$$

which entails that $|C_{X,Y|Z=z}(x,y) - xy| \leq k/4$ for every $x,y \in [0,1]$, $z \in \mathbb{R}$. Therefore,

$$C_{X,Y|Z=z}(x,y) \in \left[xy - \frac{k}{4}, xy + \frac{k}{4}\right].$$

The remainder of the proof comes from remembering that the partial copula evaluated in a point $(x,y) \in [0,1]^2$ is the expectation of a random variable (see Remark 4), which enables using the monotonicity of expectation: property E2 of section 3.3 of James [1981] readily entails that, if a random variable belongs almost surely to an interval, then the same happens to its expectation. Therefore,

$$xy - \frac{k}{4} \leq C_{X,Y;Z}(x,y) \leq xy + \frac{k}{4},$$

for every $x, y \in [0,1]$. Therefore, $|C_{X,Y;Z}(x,y) - xy| \leq k/4$, which entails that

$$D(U_X, U_Y) = 4 \sup_{x,y \in [0,1]} (|C_{X,Y}(x,y) - xy|) \leq k.$$

$\square$

The last result can be intuitively interpreted in the following way: regardless of simplifying assumption holding or not, if all the conditional copulas are distant from independence at most k, then the partial copula is distant from independence at most k too. This entails that, if all conditional copulas are close to the independence case, then the partial copula is necessarily close to independence.

Furthermore, it is possible to also bound Spearman's correlation coefficient (here denoted by the letter ρ) in the partial copula, provided an upper bound for the KDD of the conditional copulas is also known (it is worth remembering that Spearman's correlation is also a functional of the copula, see Fredricks and Nelsen [2007]).

Proposition 6. *Let X, Y, Z be jointly continuous random variables. If $(D(X, Y|Z = z)) \leq k$ for all $z \in \mathbb{R}$, then*

$$\max(-1, -3k) \leq \rho(X, Y; Z) \leq \min(1, 3k).$$

Proof. The proof of Proposition 5 entails that, if $(D(X, Y|Z = z)) \leq k$ for all $z \in \mathbb{R}$, then

$$xy - \frac{k}{4} \leq C_{X,Y;Z}(x,y) \leq xy + \frac{k}{4},$$

which entails that

$$\int_0^1 \int_0^1 C_{X,Y;Z}(x,y)dxdy \in \left[\frac{1-k}{4}, \frac{1+k}{4}\right].$$

Theorem 5.1.6 in Nelsen [2006] states that, given a copula C, its Spearman's correlation can be represented as $12 \int_0^1 \int_0^1 C_{X,Y;Z}(x,y)dxdy - 3$. Applying this representation in the partial copula, it holds that

$$-3k \leq \rho(X, Y; Z) \leq 3k,$$

as we wished to prove. The final bound in the proposition is obtained taking into account that Spearman's coefficient belongs to the range $[-1, 1]$ [see Theorem 5.1.9 in Nelsen, 2006]. $\square$

An even better bound can be obtained for the Kendall's tau correlation coefficient (denoted by τ), which is also a functional of the copula [Nelsen, 2006] - see next Proposition.

Proposition 7. *Let X, Y, Z be jointly continuous random variables. If $(D(X, Y|Z = z)) \leq k$ for all $z \in \mathbb{R}$, then*

$$\max(-1, -2k) \leq \tau(X, Y; Z) \leq \min(1, 2k).$$

Proof. We have seen in the proof of Proposition 5 that, if $(D(X, Y|Z = z)) \leq k$ for all $z \in \mathbb{R}$, then

$$xy - \frac{k}{4} \leq C_{X,Y;Z}(x,y) \leq xy + \frac{k}{4}.$$

Therefore, inserting U_X and U_Y as arguments, we have

$$U_X U_Y - \frac{k}{4} \leq C_{X,Y;Z}(U_X, U_Y) \leq U_X U_Y + \frac{k}{4}.$$

The monotonicity of expectation [property E2 of section 3.3 of James, 1981] entails that

$$E[U_XU_Y] - \frac{k}{4} \leq E[C_{X,Y;Z}(U_X, U_Y)] \leq E[U_XU_Y] + \frac{k}{4}.$$

The Equation 5.1.15b from Theorem 5.1.6 in Nelsen [2006] entails that $\rho(X, Y; Z) = 12E[U_XU_Y] - 3$, which means that

$$\frac{\rho(X, Y; Z) + 3}{12} - \frac{k}{4} \leq E[C_{X,Y;Z}(U_X, U_Y)] \leq \frac{\rho(X, Y; Z) + 3}{12} + \frac{k}{4}.$$

The final argument comes from remembering that, according to Theorem 5.1.3 in Nelsen [2006], Kendall's tau correlation coefficient can be represented as $4E[C_{X,Y;Z}(U_X, U_Y)] - 1$, which leads us to the conclusion that

$$\frac{\rho(X, Y; Z) + 3}{3} - k - 1 \leq \tau(X, Y; Z) \leq \frac{\rho(X, Y; Z) + 3}{3} + k - 1,$$

and, from the inequality $-3k \leq \rho(X, Y; Z) \leq 3k$, it follows that

$$-2k \leq \tau(X, Y; Z) \leq 2k.$$

As in the previous proof, the final bounds are obtained using the fact that Kendall's coefficient belongs to the range $[-1, 1]$ [see Theorem 5.1.9 in Nelsen, 2006]. $\square$

Additionally, the connection between the correlation in the conditional copulas and in the partial copula can be summarized in the following theorem.

Proposition 8. *Let X, Y, Z be jointly continuous random variables. Then,*

$$\rho(X, Y; Z) = \int_{-\infty}^{\infty} \rho(X, Y|Z = z) f_Z(z) dz.$$

Proof. Applying Equation 5.1.15c of Nelsen [2006] in the partial copula, we obtain

$$\rho(X, Y; Z) = 12 \int_0^1 \int_0^1 C_{X,Y;Z}(x, y) dx dy - 3.$$

Now using representation from Proposition 3, we obtain

$$\begin{aligned}
\rho(X, Y; Z) &= 12 \int_0^1 \int_0^1 \int_{-\infty}^{\infty} C_{X,Y|Z=z}(x, y) f_Z(z) dz dx dy - 3 \\
&= 12 \int_{-\infty}^{\infty} \left(\int_0^1 \int_0^1 C_{X,Y|Z=z}(x, y) dx dy \right) f_Z(z) dz - 3 \\
&= 12 \int_{-\infty}^{\infty} \left(\frac{\rho(X, Y|Z = z) + 3}{12} \right) f_Z(z) dz - 3 \\
&= \int_{-\infty}^{\infty} \rho(X, Y|Z = z) f_Z(z) dz.
\end{aligned}$$

$\square$

That is, the Spearman's correlation induced by the partial copula is the expected conditional Spearman's correlation - this greatly clarifies the role of conditional correlations in determining partial correlation. Finally, a consequence of this result in a family of copulas studied in Rodríguez-Lallena and Úbeda-Flores [2004] is illustrated in the following example.

Example 2. *Rodríguez-Lallena and Úbeda-Flores [2004] studied a nonparametric family of copulas where the copula of two random variables X', Y' can be written as $C_{X',Y'}(x, y) = xy + f(x)g(y)$. Now, consider a case where all the conditional copulas of X, Y given Z can be written as*

$$C_{X,Y|Z=z}(x, y) = xy + f(x, z)g(y, z),$$

which means that all the conditional copulas belong to the family of copulas studied in Rodríguez-Lallena and Úbeda-Flores [2004]. Komelj and Perman [2010] showed that all copulas in this family have Spearman's correlation coefficient belonging to $[-3/4, 3/4]$. Therefore, applying Proposition 8 and monotonicity of expectation [property E2 of section 3.3 of James, 1981], it is straightforward to see that

$$-\frac{3}{4} \leq \rho(X, Y; Z) \leq \frac{3}{4}.$$

Therefore, the fact that all the conditional copulas belong to the family of copulas studied in Rodríguez-Lallena and Úbeda-Flores [2004] entails that the Spearman's correlation coefficient in the partial copula is also bounded by $[-3/4, 3/4]$.

4 Simulation Study

The theoretical results in Section 3 establish that the partial copula $C_{X,Y;Z}$ captures the dependence between X and Y after removing the effect of Z. In particular, Proposition 8 shows that the Spearman correlation induced by the partial copula equals the expected conditional correlation, while Proposition 4 establishes that the sign of quadrant dependence is preserved when it is constant across all conditional copulas. In this section, we conduct a simulation study to illustrate these results and to demonstrate how the partial copula can reveal the true conditional association structure when marginal correlations are confounded. All simulations were performed in R using the `rvinecopulib` package [Nagler and Vatter, 2025]. The manuscript and all analyses are fully reproducible through the `rix` package [Rodrigues and Baumann, 2025], and the source code and environment files are available in the accompanying GitHub repository: https://github.com/felipelfv/copula_project.

4.1 Design

We employ a C-vine copula with Z as the root node, which decomposes the joint distribution of (X, Y, Z) through the pair copulas $C_{X,Z}$, $C_{Y,Z}$, and $C_{X,Y|Z}$. For each configuration, we generate a single sample of $n = 5000$ observations $(X_i, Y_i, Z_i)_{i=1}^n$ and compute the partial copula pseudo-observations via h-functions,

$$U_{X,i} = h_2(X_i, Z_i; C_{X,Z}) \quad \text{and} \quad U_{Y,i} = h_2(Y_i, Z_i; C_{Y,Z}),$$

where $h_2(u, v; C) = \partial C(u, v)/\partial v$ is the conditional distribution function of the first argument given the second.

To assess how confounding and conditional dependence interact, we compare the marginal and partial rank correlations. Specifically, we compute Spearman's ρ and Kendall's τ for both the original pair (X, Y) and the partial copula pair (U_X, U_Y). A discrepancy between these quantities indicates confounding: dependence between X and Y that is induced by Z rather than by direct conditional association.

We consider eleven scenarios, summarized in Table 1. The triplet $(s_{XZ}, s_{YZ}, s_{XY|Z})$ indicates the sign of dependence in each pair copula: + for positive, − for negative, and ind for independence. Scenarios 1–10 satisfy the simplifying assumption, so that $C_{X,Y|Z=z}$ does not vary with z and Corollary 1 applies. Scenario 11 violates the simplifying assumption by letting the conditional copula parameter vary as a function of Z through specifications $\theta(z) = \exp(z)$, $\theta(z) = -\exp(z)$, and $\theta(z) = 1-2z$; the last of these illustrates Remark 3, where positive and negative conditional associations cancel out in expectation.

Across scenarios, we consider multiple copula families (Frank, Gumbel, Clayton, and Gaussian) over a range of parameter values to ensure robustness of the findings. Negative dependence for Gumbel and Clayton copulas is achieved via 90-degree rotation.

Table 1: Summary of simulation scenarios. The triplet $(s_{XZ}, s_{YZ}, s_{XY|Z})$ denotes the sign of dependence in each pair copula: + positive, − negative, ind independence, and · varies across configurations.

Scenario	Structure	Description
1	$(+,+,+)$	Positive confounding, positive conditional
2	$(+,+,\text{ind})$	Positive confounding, conditional indep.
3	$(-,-,\text{ind})$	Negative confounding, conditional indep.
4	$(+,-,\text{ind})$	Opposite confounding, conditional indep.
5	$(+,+,-)$	Positive confounding, negative conditional
6	$(-,-,+)$	Negative confounding, positive conditional
7	$(-,-,-)$	Negative confounding, negative conditional
8	$(+,-,+)$	Opposite confounding, positive conditional
9	$(+,-,-)$	Opposite confounding, negative conditional
10	$(\text{ind},\text{ind},\cdot)$	No confounding
11	—	Simplifying assumption violated

4.2 Results

Table 2 in the Appendix reports Spearman's ρ and Kendall's τ for both the marginal pair (X, Y) and the partial copula pair (U_X, U_Y) across all scenarios. The results confirm the theoretical predictions. In conditional independence scenarios (2, 3, and 4), the partial correlations are negligible while marginal correlations can be substantial, demonstrating that the partial copula successfully removes confounding. In scenarios where confounding reinforces conditional dependence (1, 6, and 9), the partial correlations correctly recover the conditional association, which is attenuated relative to the marginal correlations. Notably, scenarios 5, 7, and 8 exhibit Simpson's paradox: marginal and partial correlations have opposite signs. In these scenarios, the confounding effect opposes the conditional dependence — for instance, in scenario 7 $(-,-,-)$, the product of two negative marginal dependencies induces a positive spurious correlation that counteracts the negative conditional dependence. This illustrates how the partial copula can reveal conditional associations that are masked or reversed by confounding. Finally, Scenario 11 examines violations of the simplifying assumption. In cases 11a and 11b, where $\theta(z) = \exp(z)$ and $\theta(z) = -\exp(z)$ respectively, the sign of conditional dependence remains constant across all values of Z, and the partial correlations closely match the marginal correlations, consistent with Proposition 4. In contrast, case 11c deserves particular attention. Here the conditional copula parameter $\theta(z) = 1 - 2z$ is a Gaussian copula whose correlation switches sign at $z = 0.5$: for $z < 0.5$ the conditional dependence is positive, while for $z > 0.5$ it becomes negative. As a result, the partial copula — which averages over all conditional copulas — yields near-zero partial correlation despite strong conditional dependence for almost all values of Z, confirming Remark 3. Figure 1 illustrates this by displaying the conditional scatter plots for $Z < 0.5$ and $Z > 0.5$ alongside the partial copula: the opposing dependence structures are clearly visible in the conditional panels but cancel out in the partial copula. This highlights a fundamental limitation: the partial copula captures average conditional dependence and may not be the best tool when one is interested in local, z-specific dependence.

5 Discussion

In this paper, we discussed the notion of partial copula, originally introduced in Bergsma [2004] to test conditional independence. This test was introduced upon the observation that, under conditional independence, the variables $F_{X|Z}(X|Z)$ and $F_{Y|Z}(Y|Z)$, usually known as Rosenblatt transformations, are independent. Although we recognize the usefulness of the partial copula as a tool to test conditional independence, our main focus has been to highlight its broader applicability as a general representation of conditional association, which has not been well explored previously in the literature, to the best of our knowledge. The starting point of this paper is the observation that the variables $F_{X|Z}(X|Z)$ and $F_{Y|Z}(Y|Z)$ are uniformly distributed in $[0, 1]$, which means that their joint cumulative distribution function is a copula, which enables studying several dependence concepts constructed in the copula literature, such as Spearman's and Kendall's coefficients.

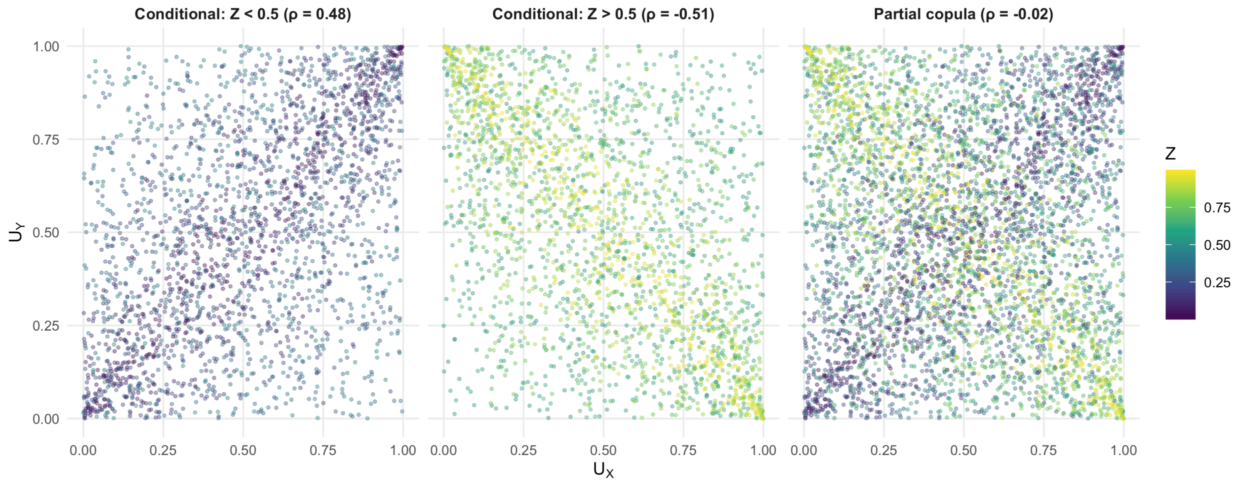

Figure 1: Scenario 11c: Gaussian copula with $\theta(z) = 1 - 2z$. Left: conditional scatter for $Z < 0.5$ (positive dependence). Center: conditional scatter for $Z > 0.5$ (negative dependence). Right: partial copula averaging over all Z. Color gradient represents Z; ρ denotes Spearman's correlation.

First, we showed in Proposition 3 and Remark 2 that the partial copula of (X, Y) adjusted by Z may be viewed as the joint distribution of (X, Y) under the assumption that X and Y are both independent of Z (marginally) but preserving their conditional copula. This interpretation clarifies the role of the partial copula as a general representation of conditional association, since it captures the dependence between X and Y after removing the effect of Z. Second, we established several properties of the partial copula, including bounds for Spearman's and Kendall's coefficients in the partial copula when the maximum value of the Kolmogorov distance dependence in the conditional copulas is known (Propositions 6 and 7), the connection between conditional and partial correlations (Proposition 8), and a characterization of the partial copula when the conditional copulas belong to a specific parametric family (Proposition 9). At an intuitive level, those results show that the partial copula captures the properties of conditional copulas while removing the effect of Z.

In the context of causal inference, if Z is a confounding variable, the dependence between X and Y adjusted by Z - as traditionally estimated in linear models and other statistical techniques - may be interpreted as the true causal connection between the two variables. The properties proved in this paper underscore the capability of the partial copula to capture this causal connection without requiring restrictive assumptions such as linearity or the simplifying assumption used in vine copula modeling.

Finally, a simulation study was conducted to compare how the partial and the marginal copulas may be different. In scenarios presenting conditional independence (scenarios 2, 3 and 4), the partial copula is the independence copula, while the marginal copula contains the spurious association induced by Z. In scenarios where the confounding effect reinforces the conditional dependence (1, 6, and 9), the association observed in the marginal copula was stronger than in the partial copula. In scenarios where the confounding effect opposes the conditional dependence (5, 7, and 8), Simpson's paradox arises: marginal and partial correlations have opposite signs. In scenarios with no confounding (10), the marginal and the partial copula are the same. Finally, scenario 11 illustrated cases violating the simplifying assumption, which shows a fundamental difference between partial and conditional copulas: conditional copulas depend on the specification of the value of Z being conditioned on, while the partial copula does not. In this sense, the partial copula may be seen as a "summary" of the conditional copulas across all values of Z.

The diversity of scenarios simulated shows that, in several contexts, the marginal association and the average conditional association may be very different - including scenarios known as Simpson's paradox, where even the signs of correlation coefficients are different. In causal inference literature [Pearl, 2009, Pearl et al., 2016], it is well known that marginal associations may be very different from causal effects due to confounding.

The results in this paper show that the partial copula is a useful tool to estimate the average conditional association, which may be interpreted as a causal effect under standard assumptions in causal inference. From this paper, we hope to have shown both the importance of using conditional association instead of marginal association in several contexts, and the capability of the partial copula to capture this conditional association in a general way.

Some directions for future research include studying directed dependence coefficients [Azadkia and Chatterjee, 2021, Fuchs, 2024] in the context of partial copulas, and exploring the use of partial copulas in the context of causal discovery algorithms. Furthermore, further advances on the estimation of partial copulas from real data are of great interest for the scientific community, since it would allow researchers to apply the insights of this paper in empirical contexts. Bergsma [2004] discussed the use of kernel estimators to estimate the conditional CDFs; however further exploration on the properties of such estimators seems necessary. Although there is a rich literature in the estimation of conditional copulas [e.g., Veraverbeke et al., 2011, Muia et al., 2026], the estimation of partial copulas itself is a subject that requires further developments.

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Appendix

Table 2: Spearman's ρ and Kendall's τ for the marginal pair (X, Y) and the partial copula pair (U_X, U_Y) across all simulated scenarios. The Specification column reports the copula family and the parameter triplet $\theta = (\theta_{XZ}, \theta_{YZ}, \theta_{XY|Z})$; (R) indicates a 90-degree rotation and ind denotes independence.

		Marginal (X, Y)		Partial (U_X, U_Y)	
Case	Specification	ρ	τ	ρ	τ
Scenario 1: (+, +, +)					
1a	Frank, $\theta = (1, 1, 1)$	0.174	0.117	0.152	0.102
1b	Frank, $\theta = (3, 3, 1)$	0.326	0.220	0.164	0.110
1c	Frank, $\theta = (5, 5, 1)$	0.495	0.341	0.157	0.105
1d	Gumbel, $\theta = (1.5, 1.5, 1.5)$	0.614	0.442	0.479	0.336
1e	Gumbel, $\theta = (2, 2, 1.5)$	0.731	0.546	0.483	0.338
1f	Gumbel, $\theta = (3, 3, 1.5)$	0.860	0.681	0.473	0.331
1g	Clayton, $\theta = (1, 1, 1)$	0.618	0.446	0.494	0.346
1h	Clayton, $\theta = (3, 3, 1)$	0.814	0.630	0.476	0.332
1i	Clayton, $\theta = (5, 5, 1)$	0.891	0.725	0.468	0.325
Scenario 2: (+, +, ind)					
2a	Frank, $\theta = (1, 1, \text{ind})$	0.031	0.020	0.003	0.002
2b	Frank, $\theta = (3, 3, \text{ind})$	0.201	0.134	-0.004	-0.003
2c	Frank, $\theta = (5, 5, \text{ind})$	0.400	0.269	-0.028	-0.019
2d	Gumbel, $\theta = (1.5, 1.5, \text{ind})$	0.270	0.183	0.025	0.017
2e	Gumbel, $\theta = (2, 2, \text{ind})$	0.467	0.325	-0.014	-0.010
2f	Gumbel, $\theta = (3, 3, \text{ind})$	0.734	0.545	0.019	0.013
2g	Clayton, $\theta = (1, 1, \text{ind})$	0.276	0.187	0.016	0.011
2h	Clayton, $\theta = (3, 3, \text{ind})$	0.645	0.468	0.008	0.005
2i	Clayton, $\theta = (5, 5, \text{ind})$	0.795	0.608	-0.005	-0.003
Scenario 3: (-, -, ind)					
3a	Frank, $\theta = (-1, -1, \text{ind})$	0.048	0.032	0.019	0.013
3b	Frank, $\theta = (-3, -3, \text{ind})$	0.182	0.122	-0.010	-0.007
3c	Frank, $\theta = (-5, -5, \text{ind})$	0.390	0.264	-0.028	-0.019
3d	Gumbel (R), $\theta = (1.5, 1.5, \text{ind})$	0.234	0.158	-0.004	-0.003
3e	Gumbel (R), $\theta = (2, 2, \text{ind})$	0.485	0.339	-0.001	0.000
3f	Gumbel (R), $\theta = (3, 3, \text{ind})$	0.724	0.534	-0.001	-0.001
3g	Clayton (R), $\theta = (1, 1, \text{ind})$	0.259	0.175	0.005	0.004
3h	Clayton (R), $\theta = (3, 3, \text{ind})$	0.652	0.474	0.011	0.007
3i	Clayton (R), $\theta = (5, 5, \text{ind})$	0.796	0.609	0.004	0.003
Scenario 4: (+, -, ind)					
4a	Frank, $\theta = (1, -1, \text{ind})$	-0.017	-0.011	0.011	0.007
4b	Frank, $\theta = (3, -3, \text{ind})$	-0.183	-0.123	0.018	0.012
4c	Frank, $\theta = (5, -5, \text{ind})$	-0.424	-0.287	-0.006	-0.004
4d	Gaussian, $\theta = (0.3, -0.3, \text{ind})$	-0.119	-0.080	-0.036	-0.024
4e	Gaussian, $\theta = (0.5, -0.5, \text{ind})$	-0.250	-0.168	0.002	0.001
4f	Gaussian, $\theta = (0.7, -0.7, \text{ind})$	-0.477	-0.330	-0.016	-0.011
Scenario 5: (+, +, -)					
5a	Frank, $\theta = (1, 1, -1)$	-0.156	-0.105	-0.190	-0.128
5b	Frank, $\theta = (1, 1, -3)$	-0.410	-0.279	-0.452	-0.309
5c	Frank, $\theta = (1, 1, -5)$	-0.598	-0.419	-0.648	-0.460
5d	Gaussian, $\theta = (0.2, 0.2, -0.2)$	-0.147	-0.098	-0.190	-0.128
5e	Gaussian, $\theta = (0.2, 0.2, -0.5)$	-0.406	-0.278	-0.467	-0.322
5f	Gaussian, $\theta = (0.2, 0.2, -0.7)$	-0.638	-0.456	-0.699	-0.508
5g	Frank, $\theta = (5, 5, -1)$	0.320	0.214	-0.170	-0.114
5h	Frank, $\theta = (5, 5, -0.5)$	0.367	0.247	-0.095	-0.064

Table 2: Spearman's ρ and Kendall's τ for the marginal pair (X, Y) and the partial copula pair (U_X, U_Y) across all simulated scenarios. The Specification column reports the copula family and the parameter triplet $\theta = (\theta_{XZ}, \theta_{YZ}, \theta_{XY|Z})$; (R) indicates a 90-degree rotation and ind denotes independence. *(continued)*

		Marginal (X, Y)		Partial (U_X, U_Y)	
Case	Specification	ρ	τ	ρ	τ
5i	Gaussian, θ = (0.7, 0.7, -0.2)	0.386	0.263	-0.202	-0.136
5j	Gaussian, θ = (0.7, 0.7, -0.1)	0.425	0.292	-0.090	-0.060
Scenario 6: (-, -, +)					
6a	Frank, θ = (-1, -1, 1)	0.186	0.124	0.166	0.110
6b	Frank, θ = (-3, -3, 1)	0.332	0.226	0.184	0.123
6c	Frank, θ = (-5, -5, 1)	0.487	0.336	0.142	0.095
6d	Gumbel (R), θ = (1.5, 1.5, 1.5)	0.614	0.441	0.485	0.340
6e	Gumbel (R), θ = (2, 2, 1.5)	0.728	0.541	0.465	0.325
6f	Gumbel (R), θ = (3, 3, 1.5)	0.872	0.692	0.484	0.340
6g	Clayton (R), θ = (1, 1, 1)	0.615	0.445	0.479	0.335
6h	Clayton (R), θ = (3, 3, 1)	0.813	0.630	0.486	0.339
6i	Clayton (R), θ = (5, 5, 1)	0.896	0.730	0.473	0.329
Scenario 7: (-, -, -)					
7a	Frank, θ = (-1, -1, -1)	-0.136	-0.091	-0.165	-0.110
7b	Frank, θ = (-1, -1, -3)	-0.417	-0.285	-0.459	-0.316
7c	Frank, θ = (-1, -1, -5)	-0.586	-0.410	-0.636	-0.451
7d	Gumbel (R), θ = (1.5, 1.5, 1.5)	-0.116	-0.080	-0.479	-0.335
7e	Gumbel (R), θ = (1.5, 1.5, 2)	-0.225	-0.157	-0.670	-0.489
7f	Gumbel (R), θ = (1.5, 1.5, 3)	-0.380	-0.269	-0.855	-0.674
7g	Clayton (R), θ = (1, 1, 1)	-0.050	-0.038	-0.467	-0.325
7h	Clayton (R), θ = (1, 1, 3)	-0.293	-0.215	-0.790	-0.603
7i	Clayton (R), θ = (1, 1, 5)	-0.343	-0.257	-0.883	-0.712
Scenario 8: (+, -, +)					
8a	Frank, θ = (1, -1, 1)	0.119	0.079	0.147	0.098
8b	Frank, θ = (3, -3, 1)	-0.096	-0.063	0.144	0.096
8c	Frank, θ = (5, -5, 1)	-0.322	-0.214	0.159	0.106
8d	Gaussian, θ = (0.2, -0.2, 0.2)	0.134	0.090	0.183	0.123
8e	Gaussian, θ = (0.5, -0.5, 0.2)	-0.091	-0.061	0.192	0.129
8f	Gaussian, θ = (0.7, -0.7, 0.2)	-0.383	-0.261	0.186	0.124
Scenario 9: (+, -, -)					
9a	Frank, θ = (1, -1, -1)	-0.191	-0.128	-0.171	-0.114
9b	Frank, θ = (3, -3, -1)	-0.360	-0.243	-0.187	-0.124
9c	Frank, θ = (5, -5, -1)	-0.497	-0.342	-0.156	-0.104
9d	Gaussian, θ = (0.2, -0.2, -0.2)	-0.224	-0.151	-0.197	-0.133
9e	Gaussian, θ = (0.5, -0.5, -0.2)	-0.387	-0.264	-0.186	-0.124
9f	Gaussian, θ = (0.7, -0.7, -0.2)	-0.574	-0.404	-0.197	-0.132
Scenario 10: (ind, ind, ·)					
10a	Frank, θ = (ind, ind, -1)	-0.179	-0.120	-0.179	-0.120
10b	Frank, θ = (ind, ind, 1)	0.165	0.110	0.165	0.110
10c	Indep, θ = (ind, ind, ind)	-0.019	-0.013	-0.019	-0.013
Scenario 11: Simplifying assumption violated					
11a	Frank, θ(z) = exp(z)	0.272	0.183	0.259	0.174
11b	Frank, θ(z) = -exp(z)	-0.253	-0.170	-0.289	-0.195
11c	Gaussian, θ(z) = 1-2z	0.006	0.003	-0.020	-0.014